

HYDROFARMS AGRI SOLUTIONS

BURIED UTILITIES, PIPE BEDDING,
TESTING AND REINSTATEMENT

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-CIV-004
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Prepared by	Hydrofarms Agri Solutions
Issue date	15 August 2026
Parent methods	HF-GH01-MST-GEN-002, HF-GH01-MST-CIV-001 and HF-GH01-MST-CIV-003

THREE INDEPENDENT HYDRAULIC CIRCUITS
The eastern production, western production and nursery circuits shall remain hydraulically independent from their operating tanks to their table groups and back to the same tanks. Cross-connections are prohibited unless shown on an approved process drawing.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor coordination and approval status.
Construction manager	Work-front sequence, interfaces, access and daily release.
Civil engineer	Trench, bedding, thrust restraint, backfill and reinstatement.
MEP/process engineer	Pipe class, route, hydraulic function, fittings, tests and interfaces.
Surveyor	Centrelines, tee chainages, inverts, falls and as-built coordinates.
QA/QC engineer	ITP, material traceability, jointing, tests and concealed-work records.
HSE manager	Excavation permit, access, plant segregation and pressure-test exclusion.
Client/consultant	Hold/witness inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method controls setting out, trenching, bedding, jointing, testing, survey, backfilling and reinstatement for the buried NFT feed and gravity-return networks. It converts the approved CAD route measurements into a field sequence that protects hydraulic performance and prevents concealed defects.

2. Scope and Package Boundary

The buried-utilities work package comprises the following activities and interfaces:

- Three independent pressurised 3-inch uPVC PN16 feed networks from the operating-tank pump interfaces to the table takeoffs.
- Two production gravity-return systems using 4-inch collection branches into 6-inch mains, plus one independent 4-inch nursery return.
- Trenches, selected bedding/surround, sleeves, thrust restraint, markers, valve boxes, cleanouts, testing and as-built survey.
- Watertight tank-room penetrations coordinated with HF-GH01-MST-CIV-003 and table riser/downpipe interfaces.
- Backfill, density testing and reinstatement of concrete/interlock/groundcover interfaces disturbed by the work.

PACKAGE EXCLUSIONS

Pumps, tank nozzles, on-table 1-inch/3/4-inch manifolds, NFT hoses, electrical cables and control wiring are outside this buried-pipe package. The project has no buried electrical cable routes; electrical distribution remains above ground on galvanized supports.

3. Governing Information

Priority	Controlled information
1	AFC coordinated buried-services plan, longitudinal profiles and tank-room opening schedule
2	Approved process flow diagram and pump/tank/table interface details
3	Approved pipe/fitting manufacturer installation and pressure-test instructions
4	HF-GH01-MST-CIV-001 excavation/backfill and HF-GH01-MST-CIV-003 room penetrations
5	Approved material submittals, solvent/gasket compatibility and ITP
6	Survey control and setting-out releases under HF-GH01-MST-GEN-002

PROFILE CONTROL

All route lengths below are adopted centreline measurements. Gravity-return inverts and falls shall follow the approved AFC profiles. A return pipe shall never be installed with reverse fall, local high point or reduced clear bore.

4. Pipe System Schedule

System	Material/class	Nominal size	Function	Joining
Production and nursery feed	uPVC pressure PN16	3 inch	Pressurised nutrient feed	Compatible solvent-weld socket system
Production return branches	Heavy-duty uPVC drainage SN8	4 inch	Gravity collection	Gasketed sockets / approved system
Production return mains	Heavy-duty uPVC drainage SN8	6 inch	Gravity return to 10 m³ tanks	Gasketed sockets / approved system
Nursery return	Heavy-duty uPVC drainage SN8	4 inch	Gravity return to 3 m³ tank	Gasketed sockets / approved system

Critical setting-out separations

- The clear surface-to-surface separation between the eastern and western production feed mains is 2.235 m; it is not a centreline dimension.
- The clear surface-to-surface separation between the nursery pump discharge pipe and the western production pump discharge pipe is 5.741 m.
- All tee chainages, elbow-leg lengths and route lengths in the schedules are measured centre-to-centre unless explicitly described as surface-to-surface.

5. Approved Feed Routes

Code	Route	Direction	Length (m)	Setting-out control
EF-00	East production pump main	North	28.2633	Tees at 1.7000, 9.7633, 20.2783 and 28.2633 m
EF-01	East production branch 1	East	61.1330	From tee centre to group end
EF-02	East production branch 2	East	61.1330	From tee centre to group end
EF-03	East production branch 3	East	61.1330	From tee centre to group end
EF-04	East production branch 4	East	61.1330	From tee centre to group end
WF-00	West production pump main	North	28.2633	Same tee chainages as east circuit
WF-01	West production branch 1	West	43.0575	From tee centre to group end
WF-02	West production branch 2	West	43.0575	From tee centre to group end
WF-03	West production branch 3	West	61.1330	From tee centre to group end
WF-04	West production branch 4	West	61.1330	From tee centre to group end
NF-01	Nursery pump entry leg	North	1.4400	Then long 90-degree turn west
NF-02	Nursery west main / branch 1	West	55.3059	Branch-2 tee at 39.248 m from first elbow centre
NF-03	Nursery branch-2 stem	North	8.0370	From tee centre, then turn west
NF-04	Nursery branch-2 terminal	West	16.0899	Removable flushing end

6. Approved Return Routes

Code	Route	Size	Length (m)	Setting-out / flow control
ER-M	East production return main	6 inch	39.1911	South to tank; 4-inch branches at 20.456 and 39.000 m
ER-01	East return branch 1	4 inch	61.1330	45-degree reducing wye into ER-M
ER-02	East return branch 2	4 inch	61.1330	45-degree reducing wye into ER-M
WR-M	West production return main	6 inch	39.1911	South to tank; same collection chainages
WR-01	West return branch 1	4 inch	43.3350	45-degree reducing wye into WR-M
WR-02	West return branch 2	4 inch	61.1330	45-degree reducing wye into WR-M
NR-01	Nursery tank-side leg	4 inch	4.3000	Measured opposite flow; actual flow toward tank
NR-02	Nursery second leg	4 inch	37.9000	Long-radius turn at end
NR-03	Nursery third leg	4 inch	16.9000	Long-radius turn at end
NR-04	Nursery remote terminal	4 inch	16.9000	Cleanout at remote end

7. Installed and Procurement Quantities

Network	Installed centreline (m)	Procurement quantity	Basis
East production feed, 3 inch	272.7953	Included in 654 m	CAD centreline
West production feed, 3 inch	236.6443	Included in 654 m	CAD centreline
Nursery feed, 3 inch	80.8728	Included in 654 m	CAD centreline
Feed room-entry/riser allowance	3.0000	Included in 654 m	1 m per circuit
3-inch feed total	593.3124	654 m = 109 x 6 m	10% allowance, full lengths
Production 4-inch branches	226.7340	Included in 336 m	East 122.266 + west 104.468
Nursery 4-inch return + entry	77.0000	Included in 336 m	76 m route + 1 m entry
4-inch return total	303.7340	336 m = 56 x 6 m	10% allowance, full lengths
Production 6-inch mains + entries	80.3822	90 m = 15 x 6 m	78.3822 m route + 2 m entries

QUANTITY CONTROL

The installed lengths are routing quantities, not payment quantities for completed work. Site measurement, approved shop drawings and signed as-built survey govern final valuation. Procurement allowances shall not be added a second time in the BOQ.

8. Buried Fittings and Interfaces

Item	Installed qty	Purchase qty	Application
3-inch equal PN16 pressure tee	9	10	Eight production branches plus nursery branch
3-inch PN16 long 90-degree elbow	8	9	Direction changes and room/riser transitions
3-inch full-bore double-union valve	10	11	Feed-group isolation
3-inch removable flushing end cap	12	14	Main and branch terminals
3-inch pump-discharge interface kit	3 sets	4 sets	Union/flange, flexible connector and gauge
3-inch watertight wall sleeve	3	4	Operating-tank room penetrations
3 x 1-inch reducing table takeoff	84	93	78 production + 6 nursery tables
6 x 6 x 4-inch 45-degree reducing wye	4	5	Two collection branches per production circuit
4 x 4 x 3-inch 45-degree reducing wye	84	93	Table downpipe into buried return
6-inch long-radius 90-degree elbow	4	5	Production room entries/final turns
4-inch long-radius 90-degree elbow	5	6	Nursery route and coordination turns
4-inch removable cleanout	5	6	Remote ends/changes of direction
6-inch cleanout/end cap	2	3	One per production return main
6-inch production tank return-entry kit	2 sets	2 sets	Sleeve, seal, flexible/dismantling joint and valve
4-inch nursery tank return-entry kit	1 set	1 set	Sleeve, seal, flexible/dismantling joint and valve

Emergency repair couplings and spare gasket rings are maintenance spares and shall be listed once in the spare-parts package, not as installed civil quantities.

9. Materials and Storage

- Supply pipes and fittings from compatible approved systems with permanent markings showing manufacturer, class, size and batch.
- Store pipes on level padded supports, shaded from prolonged sunlight, restrained from rolling and protected from contamination or impact.
- Keep solvent cement, cleaner and gaskets in original containers within temperature and shelf-life limits. Reject expired or contaminated products.
- Cap all open pipe ends during transport, storage and installation. No soil, concrete washout, insects or debris may enter the network.
- Use only manufacturer-approved lubricant on gasketed return joints. Petroleum grease and unapproved sealants are prohibited.

10. Detailed Installation Method

10.1 Route release and trench excavation

- Set out centrelines, tees, elbows, valve boxes, cleanouts, sleeves and tank entries from the approved control network.
- Confirm no conflict with greenhouse columns, table legs, interlock edges, drains, tank rooms or equipment foundations before excavation.
- Excavate in controlled lengths under HF-GH01-MST-CIV-001. Keep spoil and plant outside the assessed trench-edge zone.
- Trim the trench bottom without disturbing formation. Remove soft spots and replace them by the approved earthwork repair.

10.2 Bedding and pipe support

- Place at least 150 mm compacted selected bedding, or the greater manufacturer/AFC requirement, across the full trench width.
- Shape bedding for continuous barrel support while leaving socket recesses so joints do not carry point load.
- Use surveyed grade pins for gravity returns. Do not correct grade by stones, timber pieces or isolated hard packing.

10.3 Pressure-feed jointing

- Cut square, deburr and dry-fit 3-inch PN16 pipe. Mark insertion depth and verify fitting orientation before applying solvent.
- Clean, prime where required, apply approved solvent uniformly and assemble without twisting beyond the manufacturer method.
- Hold each joint for the specified initial set and observe full curing time before movement or pressure testing.
- Install tees and long bends to the route schedule. Provide engineered thrust restraint where the manufacturer/design requires it.

10.4 Gravity-return jointing

- Clean sockets and gaskets, check gasket seating, lubricate with the approved product and insert to the witness mark.
- Orient 45-degree reducing wyes in the direction of flow. Square tees or opposing entries that obstruct gravity flow are prohibited.
- Maintain the approved invert continuously through 4-inch branches, 6-inch mains, elbows, sleeves and tank entry kits.
- Provide accessible cleanouts at scheduled remote ends and changes of direction; do not bury a cleanout beneath a permanent table support.

10.5 Tank-room and table interfaces

- Pass pipes through cast sleeves/puddle-flange openings without transferring soil or structural loads to plastic tank nozzles.
- Use flexible/dismantling connections at tank entries and maintain valve/union access for removal and maintenance.
- Terminate buried feed takeoffs in accessible boxes before the 1-inch table riser. Support risers independently from buried pipe sockets.
- Connect table return downpipes through swept 45-degree reducing wyes and maintain inspection access before closure.

10.6 Surround, markers and preliminary backfill

- Place selected surround evenly by hand under haunches and both sides, then continue to at least 300 mm above pipe crown.
- Compact without displacement or ovalisation. Heavy compactors shall not operate until the required protective cover is achieved.
- Install detectable identification/marker tape above each route and permanent route/valve labels at accessible points.
- Leave joints, fittings and penetrations exposed until the relevant visual and test hold points are released.

11. Testing, Flushing and Acceptance

11.1 Pressurised 3-inch feed networks

- Test each eastern, western and nursery circuit independently with clean water after full solvent cure and before final backfill.
- Fill slowly from the low point, vent trapped air, stabilise temperature/pressure and use calibrated gauges at suitable limits.
- Apply the approved test pressure derived from design working pressure and manufacturer instructions without exceeding PN16 component ratings.
- Maintain the specified stabilisation and test periods; inspect every exposed joint and record pressure, temperature, added water and leakage.
- After acceptance, flush at sufficient velocity to remove debris and discharge only to the approved collection/evaporation route.

11.2 Gravity 4-inch and 6-inch returns

- Survey every invert, calculate installed fall between control points and confirm there is no reverse fall or local high point.
- Conduct the approved low-head water/leakage test in isolated sections with all joints and tank penetrations visible.
- Pass a proving ball/mandrel or camera where required to confirm clear bore, alignment and absence of debris or deformation.
- Demonstrate free gravity flow from representative far table branches to the correct operating tank without cross-circuit transfer.

12. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	AFC routes, profiles and interface schedule	Document review	H	Civil/MEP/QA-QC	Work-front release
02	Pipe/fitting/solvent/gasket delivery	Visual/certificates	H	MEP/QA-QC	Material release
03	Route and conflict setting out	Survey/visual	H	Survey/Civil/MEP	Route release
04	Trench formation and bedding	Level/visual	H	Civil/QA-QC	Bedding release
05	Feed joints/fittings/thrust restraints	Visual	W	MEP/QA-QC	Jointing record
06	Return inverts, wyes and cleanouts	Survey/visual	H	MEP/Survey/Consultant	Invert schedule
07	Tank-room penetrations/interfaces	Survey/visual	H	MEP/Civil/Consultant	Opening release
08	Feed pressure test	Hydrostatic test	H	MEP/QA-QC/Consultant	Pressure report
09	Return leakage/flow test	Water/proving test	H	MEP/QA-QC/Consultant	Gravity test report
10	Flushing and cleanliness	Functional/visual	W	MEP/QA-QC	Flush record
11	As-built survey before surround closure	Survey	H	Survey/QA-QC/Client	As-built file
12	Surround, marker and first backfill	Visual/density	H	Civil/QA-QC	Concealed-work release
13	Final backfill and reinstatement	Density/level/visual	W	Civil/QA-QC	Reinstatement record
14	Dossier and system handover	Record review	R	QA-QC/Client	Turnover index

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

13. Hold Points

- HP-BU-01: Coordinated route, profile and interface schedule accepted before excavation.
- HP-BU-02: Trench formation and bedding level accepted before pipe placement.
- HP-BU-03: Gravity return inverts, wye orientation and tank penetrations accepted before surround.
- HP-BU-04: Three independent feed pressure tests accepted before final backfill.
- HP-BU-05: Gravity leakage, proving and functional-flow tests accepted before final backfill.
- HP-BU-06: As-built coordinates, inverts, fittings and photographs accepted before concealment.
- HP-BU-07: Density, reinstatement and turnover dossier accepted before operational handover.

14. Backfill and Reinstatement

- Continue approved selected backfill in loose lifts not exceeding 200 mm after concealed-work release.
- Achieve minimum 95% laboratory maximum dry density beneath paved, structural and service areas; use the earthwork ITP testing frequency.
- Reinstatement interlock with the approved sub-base, sand bed, edge restraint and finished levels without settlement strips.
- Reinstatement concrete panels with compatible thickness, reinforcement, joint layout and surface finish; do not bridge an untested trench.
- Repair groundcover interfaces without creating water traps, exposed sharp edges or unsealed penetrations.
- Survey final levels and drainage falls and keep permanent valves/cleanouts visible, labelled and accessible.

15. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Trench collapse/fall	Permit, safe slope/shoring, barriers, access and competent inspection	Medium
Plant/person interaction	Segregated route, banksman, alarms and exclusion zone	Medium
Pressure-test energy	Barricade, controlled filling, calibrated equipment and no joint adjustment under pressure	Medium
Solvent vapour/fire	SDS, ventilation, gloves/eye protection, no ignition and closed containers	Low
Manual handling	Team/mechanical lift, correct supports and controlled pipe movement	Low
Contaminated discharge	Clean test water and discharge only to approved collection/evaporation route	Low
Open pipe contamination	Caps, clean storage and end-of-shift inspection	Low

16. Records and Handover

- Material certificates, pipe/fitting batch traceability and solvent/gasket shelf-life records.
- Route release, trench/bedding inspection, jointing logs and penetration acceptance.
- Pressure, gravity leakage, proving, flushing and functional-flow test reports.
- As-built CAD/PDF drawings showing coordinates, inverts, tees, elbows, valves, cleanouts and crossings.
- Concealed-work photographs with route code and chainage before surround and before final backfill.
- Backfill lot register, density reports, surface reinstatement acceptance and NCR closeout.
- Signed circuit handover proving east production, west production and nursery return to their correct tanks.

NEXT METHOD STATEMENT

The next coordinated submission should cover reinforced-concrete floors, interlock paving, equipment foundations and final surface interfaces, using the accepted buried-utility as-built survey before any permanent floor area is closed.